AI-Driven Personalized Health Risk Assessment and Treatment Recommendation System

Tushara Spandana Tejavath
Department of Data Science
University of North Texas
Denton, TX
TusharaSpandanaTejavath@my.unt.edu

Kushal Kumar Reddy Kolli
Department of Data Science
University of North Texas
Denton, TX
KushalKumarReddyKolli@my.unt.edu

Sreeja Peraka
Department of Data Science
University of North Texas
Denton, TX
SreejaPeraka@my.unt.edu

***Abstract*—According to the British Medical Journal [1], every year 795,000 people are left permanently disabled or die due to possible misdiagnosis or late diagnosis of any critical health condition in the United States. AI tools are one of the fast-evolving inventions within the Healthcare sector. It helps to improve diagnosing diseases experience. The current study will, therefore, focus on developing a new system powered by AI to analyze the profile and reports of each patient regarding personal health information. Data extraction will be served by libraries like PyPDF2 which will be used in cases when the source contains structured text, or OCR tools such as Tesseract for cases where text comes from scanned papers or hand written notes. The extracted data will then be structured and cleaned using custom scripts which are unique for medical data and provides a personal risk assessment for people and treatment suggestions to patients by analyzing the generated reports of a full body check-up, enabling diagnosis errors to be reduced and early detection of diseases based on iterative updates of real-time data of patients. Preliminary results have portrayed that, with ranked seriousness, the system identifies risks with an accuracy of The project minimized the errors in the texts extracted from structured PDFs to 5 percent but faced 10 percent errors from scanned or handwritten ones. It is noticeable that AI-made recommendations are close to experts knowledge, including the case of anomalies and treatment. In summary, I observed stable performance by the entire system in terms of extracting medical reports, together with personalized treatment plans. The current work describes the importance of integrating AI technologies into traditional diagnosis systems to raise the percentage of risk detection.**

***Index Terms*—Artificial Intelligence, Risk, Natural Language Processing, Early Detection, Personalized treatment**

I. Introduction

Diagnosis of diseases in the domain of healthcare plays an important role in satisfactory identification of patients' medical conditions. It enables effective treatment planning and helps in reducing the suffering and mortality rates among patients. Traditional forms of diagnosis require manpower and sometimes lead to misdiagnosis, errors, and ignorance of critical signs that the human body gives, which are crucial for early detection of diseases. "Estimated that 12 million Americans are affected each year, and maybe one-third are harmed as a result. These errors may account for 40,000 to 80,000 preventable deaths annually in U.S. hospitals" [4]. Because of these minor errors, patients sometimes have to pay either with their money or their lives.

AI and Machine Learning together in healthcare have made significant advancements over the past decade. Some targets of improvement in overall health outcomes have served as cornerstones for personalized medical assistance, resource allocation, and preventive care. The system also incorporates image-based diagnostics as appropriate where in cases such as observing abnormalities in scans of the body. Also, it can perform multiple conditions at once which increases possibilities of early diagnosis hence, increases thorough and accurate results for specified health conditions.

In our project, we apply various state-of-the-art AI approaches including Generative AI and ML in various stages such as data mining with unstructured data from medical reports, and developing or recommending patient-specific treatment plans. These AI models are also useful in identifying text-based asymmetry of reports and thus improving the comprehension of health conditions. This project uses diverse, high-quality real-time datasets from EHRs and medical imaging ensuring equitable and accurate diagnostic results bridging the gap between clinical knowledge and patient understanding to better engagement with care plans.

These focus on advanced AI techniques integrated with models of personalized care. The resultant project improves diagnostic accuracy and better access to this technology, especially for underserved populations. Preliminary results reflect significant improvements in diagnostic accuracy, reduced false-positive/negative rates, and increased clinician trust in AI-assisted decisions.

II. AI in diagnosis

A. Diagnostic Errors and Their Impact on Healthcare Outcomes

Newman-Toker [1] give reason with the ominous impact of diagnostic errors, showing that every year around 12 million American adults experience a diagnostic error, and between 40,000 to 80,000 die from mistakes that could have been avoided in U.S. hospitals. Following this review, for the first time, there was a revelation of the human limiting factors that always come up during traditional diagnosis, including expertise at the individual level and a propensity for several cognitive biases. Presents strong evidence of the extent and harm from diagnostic errors. These findings show the urgent

need for sophisticated systems that minimize human error. Most of the contributions that form the basis of this paper focus on quantification of errors without elaborative solutions. This study identifies AI as the main player in the fight against diagnostic error. Our concept further facilitates early diagnosis with minimal chances of mistakes while incorporating NLP and ML hence responding to the gap Newman-Toker et al have presented.

B. AI in Early Diagnosis and Personalized Treatment

Kale [2] review the integration of AI in diagnosis of Alzheimer's disease and demonstrate that AI is capable of early disease detection and its treatment in a personalized manner using machine learning models. The predictive modeling capabilities, identifying those that will enable clinicians to forecast a patient's disease progression to allow the design of a personalized care plan, are identified. It demonstrates how AI can enhance early disease diagnosis and provide personalized treatments. Predictive modeling offers various advantages with respect to resource optimization. Some challenges include biased data leading to limited generalizability of AI models and constricting data privacy that inhibits wide diffusion. Thus, our project grows the list of lethal health conditions it can cover by applying AI to unstructured outpatient information including a text of a report. This approach assists in surmounting the problem areas associated with the specificity of the data since the AI models are generic, and not unique to a particular type of health ailments.

C. AI Predictions and Human Judgment

Gondocs and Dörfler [3] explored how AI predictions interact with human decision-making in the clinical domain. Their work was focused on the support of AI decisions, rather than their replacement by human judgment, and the related trust issues in the performance of the AI Systems. The investigation promotes cooperation between humans and AI, in view of a balanced approach to the implementation of AI and provides insight into the psychological dimensions of trust in AI-driven diagnostics.

Although it does not synthesize how the degree of precision in AI impacts patient outcomes, its focus is narrow to real tools that provide support in making decisions. Our proposal falls within this ambit by embedding the AI tools in practitioner oversight and addressing the trust issues through emphasis on explainable AI for an understanding of proprietary diagnostic recommendations. The research described the work on the use of Artificial Intelligence in diagnostics related to skin diseases, based on the basic work of Dora. Their studies have concentrated on its decision support system for melanoma detection. Their research helps in understanding how human expertise interacts with AI to drive decision-making in healthcare. The traditional approaches based on health professional manual assessment supported by clinical guidelines and personal expertise base their findings from repeated tests supported by medical imaging and pathology interpreted by specialists. Clinical Decision Support Systems comprising predictive analytics form part of the conventional treatment planning. AI-based technologies usually rely on unstructured data passing through ML algorithms that may be trained on labeled datasets using NLP to extract insight. Although effective, such applications have biased diagnoses when the underlying data is not representative for diverse populations, resulting in high false positive and false negative rates and further loss of patient trust.

Dora Ferrari [3] stresses that AI is to act as a support to the medical professional; hence, AI could be used as assistants or peer colleagues during clinical decision-making. This study, although strong in many dimensions, does lack the measurement of the benefit to the patients due to the system. Limitations overcome, the integration of NLP, ML Models, and LLM brings in automation to the processing of medical reports with reduced errors and improved accuracies. This model is able to assess multiple conditions at once, prioritize the interventions by severity, and develop personalized treatment plans that match individual health profiles. It will enhance adherence and outcomes by providing adaptive, visually appealing recommendations directly to the patient, bridging the gap between clinical knowledge and patient empowerment.

D. Comparative Analysis of AI Models

The study by the Editorial Team [5] compared Claude3 and GPT-4 models concerning medical knowledge applications, namely complex diagnosis exercises. The results of these tests demonstrated the superiority of GPT-4 in certain aspects, providing more detailed insight into medical information, but both models were limited real-time adaptive tools. The present study, while underlining strengths as well as weaknesses of the state-of-the-art AI models, points to gaps in the effective handling of unstructured medical data. However, it focuses on language models alone and does not integrate other AI techniques. Besides, no insights into clinical validation of the models are provided. This study provides foundational knowledge for integrating GPT-like models into our NLP-based diagnostic framework. Our proposal fills up the gap in terms of practical implementation by addressing real-time adaptability and clinical validation.

E. Challenges in AI Diagnostics

Analytics Emerging India [6] gives an overview of AI in diagnostics, including challenges: bias, data privacy, and integration. It further stresses the transforming potential of AI in early detection and improvement of efficiency. Pros: Thorough discussion of both opportunities and limitations. Identifies actionable areas for improving AI integration in healthcare. Cons: Lack of specific case studies or technical implementations. Relevance to Proposal: Our research addresses such challenges through the incorporation of secure data practices and scalable AI systems that make for unbiased and effective diagnostics based on broad ideas discussed in this study.

F. Machine Learning in Health Risk Assessment

The work of Abhadiomhen [9]systematically explores machine learning in health risk assessment. ML played a prime role in predicting health outcomes and identifying high-risk cases. Also, it has shown the capability of ML for the improvement of strategies concerning preventive care. Pros: Presents ML as a flexible tool which may be applied on various analytics from different data sources. Offers an overview of techniques for risk stratification. Cons: This work has given very limited importance to the role of NLP in risk assessment. Difficulty in balancing the accurate identification rate with the false positive rate in most of the prediction cases. Relevance to Proposal: The approach extends these ideas by incorporating NLP with ML to automate diagnostics with actionable insights, hence overcoming the limitations on data diversity and its accuracy.

These studies, put together, form very strong grounds for the development of AI in medical diagnostics. Our proposed methodology, while addressing the challenges of bias, trust, and integration, exploits state-of-the-art AI technologies to boost the triad of accuracy, efficiency, and personalization in healthcare. This review sets the stage for the exploration of methodologies and technologies underpinning our research in subsequent sections.

III. Methodology

The concept of the project begins with the user uploading the medical report in either PDF format or an image. It then converts all these reports to text either through text to tools that can recognize written or scanned texts. Later that data is processed, but filtered from these extraneous features. The AI next processes the report it looks at for health issues or conditions and also has image recognition for scans. The necessary tests can be run simultaneously, and the system can concurrently look for various health problems. On this basis, it suggests corresponding treatments, including medications and lifestyle changes. Last but not least, the system provides a chatbot that would be able to help users address their concerns and understand their results.

A. Dataset and Data Processing

The dataset we are working with for this project is primarily medical reports that include both PDF-format text-based documents and images of scans. These reports include key aspects of a patient's status, including his or her signs and symptoms, disease, and treatment regimes. Some reports are simple narrative, others might contain illustrations like, for example, an X-ray or an MRI showing a physically subordinated problem. The data is obtained from different healthcare organizations and is then deidentify to protect the patient's identity. The dataset is big, thousands of reports are used in training the AI system while subsets are used for testing purposes.

In data processing the first action to get done in analyzing the reports is to strip the text from their body. For the text based files, the system uses PyPDF2 tool to extract the information while for the text from images it uses OCR. Cleaning involves the removal of the unwanted portion in the text including unwanted spaces or sections of the text. This ensures that only the health data is processed making it easier for the AI to analyze and deliver value.

TESTS	RESULT	FLAG	UNITS	REFERENCE INTERVAL
Comp. Metabolic Panel (14)				
Glucose	70		mg/dL	65 - 99
BUN	15		mg/dL	6 - 24
Creatinine	1.00		mg/dL	0.76 - 1.27
eGFR If NonAfricn Am	92		mL/min/1.73	>59
eGFR If Africn Am	106		mL/min/1.73	>59
BUN/Creatinine Ratio	15			9 - 20
Sodium	139		mmol/L	134 - 144
Potassium	4.0		mmol/L	3.5 - 5.2
Chloride	100		mmol/L	96 - 106
Carbon Dioxide, Total	25		mmol/L	20 - 29
Calcium	9.0		mg/dL	8.7 - 10.2
Protein, Total	6.5		g/dL	6.0 - 8.5
Albumin	4.0		g/dL	3.5 - 5.5
Globulin, Total	2.5		g/dL	1.5 - 4.5
A/G Ratio	1.6			1.2 - 2.2
Bilirubin, Total	1.0		mg/dL	0.0 - 1.2
Alkaline Phosphatase	45		IU/L	39 - 117
AST (SGOT)	40		IU/L	0 - 40
ALT (SGPT)	40		IU/L	0 - 44

Fig. 1. Sample Data of the medical report

For the purpose of this project, the data is selected and arranged in a structured format to enhance its quality. Issues with data such as age and disease are checked and reciprocal so are measurements like age, both genders and age differences. This is done to filter out all the noise so that the relevant information for the AI algorithm to make its analysis of health is properly shown. For better performance, synthetic data is sometimes used and let's supplement for different age, gender and health issues in order to serve all the people using the system with an AI.

The second phase of the AI is when it begins to analyze the data once the data has been cleaned and sorted. For text, the Generative AI model focuses on reading through the report to detect any possible signs of any kinds of health problems or distortions. In medical images, problems such as fractures or tumors are detected using Convolutional Neural Networks (CNNs). These movements then give certain treatment or contemplated changes in lifestyle depending on what the system finds in the text and images. It permits to offer a whole picture of the user's health state to the system.

B. System Architecture

The system begins from the user interface, where medical reports are uploaded in either texual or graphical form. Such reports are writings such symptoms and diagnoses; images such as X-rays or MRI scans. The system then extracts the data from such reports especially focusing on documents containing textual data in PDF format or the scanned PDFs in the form of images using the PyPDF2 tool for the former and the OCR

(Tesseract) for the latter to transform it into readable data formats.

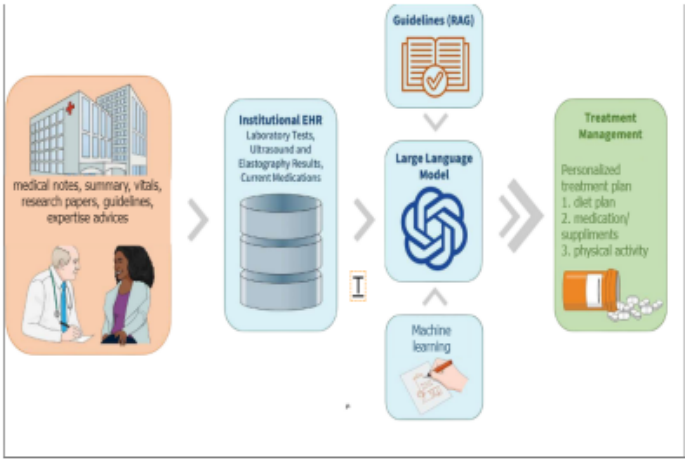

Fig. 2. Sample Data of the medical report

The system then has to preprocess the data by removing of parts which are not needed, such as headers or footnotes, of the gathered data. It makes the data accurate by normalizing elements such as the age and illnesses.. In the worst-case scenario, some data may be missing, or some records may be incomplete; in such cases synthetic data may be generated to complete data, ensuring that for any health condition, the AI is ideal.

First, the data preprocessing layer performs the cleaning, structuring, and normalization of patient data. Next, the machine-learning layer implements supervised machine-learning classifiers for the classification of numerical data. Finally, the NLP layer implements every variant of the LLMs for the analysis of text and generating insights from unstructured medical data.

After that the data is prepared and fed to the AI models to look for any health related complications. Generative AI scans the text appearing in other people's writings for any signs of illness or disease, whereas CNNs diagnose the images that reveal physical ailments. From this analysis, the system then produces treatment recommendations to suggest to the user. It is also possible to use an instant messenger chatbot that explains the answers and helps to interpret the results given. As a whole, the framework is safe as well as can be accessed with the aid of a cloud system, which guarantees privacy as well as availability to the customers.

Exploratory data analysis (EDA) and Hypotheses for the Study

1. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is performed before the application of AI on the medical data to get familiar a little bit of the data and work done before and after on it. In the case of text based reports, EDA analyzes the quality of the text that has been extracted, which fields are missing or are inconsistent, if there are any frequently mentioned diseases or symptoms. This means that irrelevant information will be reduced when using an AI.

In medical images, therefore, EDA plays the role of ensuring that the scans produced are of the best quality, something that involves checking on their resolution and format. He also examines differences in the type of images, differences in luminosity or contrast in X-rays or MRIs for example. The normal or the abnormal scans that it let AI system know that EDA is working on leveling the data so the AI system can provide a correct result in health condition.

2. Hypotheses for the study

The hypothesis for this work is that applying AI solutions for the medical report analysis enhances the accuracy of disease diagnosis and can detect diseases at an earlier stage than using other methods. With the help of recognizing textual and image data from a patient's record, the system also recognizes abnormalities and potential threats in the examination results. They also anticipate that proposed treatments produced by such AI will be more catered to the illness type and stage of the patient, enhancing general patient care.

This work assumes that the integration of such tools as the text's artificial intelligence analysis, image, and individual recommendations as the tools that will help to decrease the likelihood of diagnostic mistakes and offer faster and more accurate outcomes. If successful, the system could revolutionize communication between clinicians and patients in matters concerning their health and involve them in the discovery of any complications early enough in their development to enable provision of solutions that will help patients lead healthier lives.

3. Use Case Example

Screening Tests of Blood Tests for Early Diagnosis of Anemia. A patient comes for a normal physical examination; his/her blood sample is taken and analyzed; the analysis results go online. The system allows for parameters such as hemoglobin, red blood cell, counts, and iron pull from the blood report directly. It separates out any out-of-range values, such as low hemoglobin or low iron levels that may suggest the early discovery of anemia. Using the analysis the system suggests modifications such as changes in the diet or addition of iron products in the region. The doctor gets these suggestions and can easily advise on an appropriate treatment to help the patient avoid other complications, and proper management is done.

In a medical scenario:

- x_1: Glucose level
- x_2: Blood pressure
- β_1, β_2: Learned weights indicating how strongly these features influence the outcome.

The model calculates the probability $P(y = 1|X)$, which can predict whether a patient has a disease based on their input features.

Predictive Model: ML models predict potential conditions based on lab values, while LLMs analyze the textual context.

Output: Provides diagnostic suggestions and prioritizes interventions based on severity.

Machine Learning for Classification We use a supervised learning model for structured data analysis (e.g., lab values). The model takes patient features and maps them to a diagnosis Who classify a set of data you using a classification algorithm.

As for model training, a simplest logistic regression was used because of its simplicity and interpretability. Regarding hyperparameters of the model the learning rate and the regularization strength were chosen using Grid Search with Cross Validation = 5. Optimization was carried out using the Gradient Desent algo, and regularization strategies were used to prevent over-fitting. Python was selected as the main programming language accompanied by the Scikit-learn from the machine learning libraries, and Pandas and NumPy for data manipulation and Matplotlib for data visualization. The implementation was performed on the hardware platform by an Intel Core i7 processor with 16GB RAM.

Performance was measured in the context of model evaluation where metrics such as accuracy, precision, recall, F1 score and ROC-AUC were used. Some issues were data imbalance for which principle techniques like SMOTE for reaching synthetic samples were used while over-fitting which was checked by use of regularizations. The implementation of the logistic regression model maintained model interpretability; it is crucial in medical applications given the understanding of model behavior.

5. Data Analytics

This paper argues that in this particular project, data analytics is highly significant for transforming raw medical data into valuable information. The first step involves data scraping which involves pulling learnt data from other sources including text-base medical reports and images using python libraries including PyPDF2 and OCR. Subsequently, but before the data can be analyzed, it is "scrubbed" to strip it of any unnecessary, or noise information leaving only the pertinent health information such as the symptoms, vital signs, lab results, and diagnosis for analysis.

After that, cleaning is performed and the system analyzes the data with high attention to details, patterns and other things that look suspicious. For example, it could search for recurring references to things such as, fatigue or some altered blood values that would point towards health risks. This analysis enables early markers of diseases such as heart diseases diabetes or anemia to be spotted early and therefore conditions do not escalate. Lasty, the system applies the findings to the identified patterns to build the recommended plan of a patient. These may include suggestions on what to eat or not to eat or recommendation to take some medication or even suggestions that the patient needs to get some tests done. The continuous analysis of new data in any particular case makes the system evolve and provide more accurate and relevant results. This process helps the healthcare providers to make the right decisions and to take necessary actions early enough when it comes to handling their clients hence enhances the improvement of the results of the client.

IV. Discussion

This project centers on the way artificial intelligence shall be used in the diagnosis of diseases as well as medical reports and tests to come up with timely diagnosis of diseases and recommendation of the most appropriate treatment. The failure pattern extraction examines text-based medical reports as well as medical image data to determine core health risk patterns that the system may detect. It also provides an effective means through which medical practitioners are able to diagnose a situation that may include diabetes, heart disease or anemia amongst others, before they escalate to the next level thereby giving a good probable method through which healthcare can be taken. It also comes up with a personalized advice on the type of treatment that the patient needs depending on the health history making it more relevant for treatment plans.

Two of the main problems solved by this work are related to the management of various types of medical data – text and images. The output generality of the system in terms of text and image analysis enables the system to provide a combination of both in order to close the gap in medical information processing. Despite not being built using machine learning, employing data extraction and analysis algorithms do help improve the diagnostic ability of the AI model as well as the patient prognosis. This system does evolve as the amount of data processed increases, meaning that it provides good insight into how to efficiently and effectively make better healthcare related decisions.

V. Results

This development life cycle sub phase of this project involves the outlining of how the application is going to operate and, more specifically, how it is going to act with regards to the users of the application. It involves designing and implementing interfaces using a web-app framework called Streamlit which has functionalities such as a neatly designed Login to ensure the privacy of the users, Medical Report Analyzer page where the users can upload and get insights about the details mentioned in the PDF files as well as a Chatbot page where users can discuss about different health tips. Much like a workflow, the management of tasks is divided into sections starting with file upload, text extraction and analysis, and leading to the display of result in different sections including abnormities, conditions and recommendations respectively.

The design also incorporates a seamless incorporation of enhanced features such as text mining, analysis with the aid of artificial intelligence and practical interactive chat. To make it reliable, it has conditions for such errors as API key which may be missing, or the login details are incorrect. This is because the app has adopted a strategy whereby all the questions posed to the AI will have appropriate procedures and results in analyzing medical data outcomes. All in all, it prepares the app at this phase to be more user friendly and effective to enable healthcare providers and patients to required level.

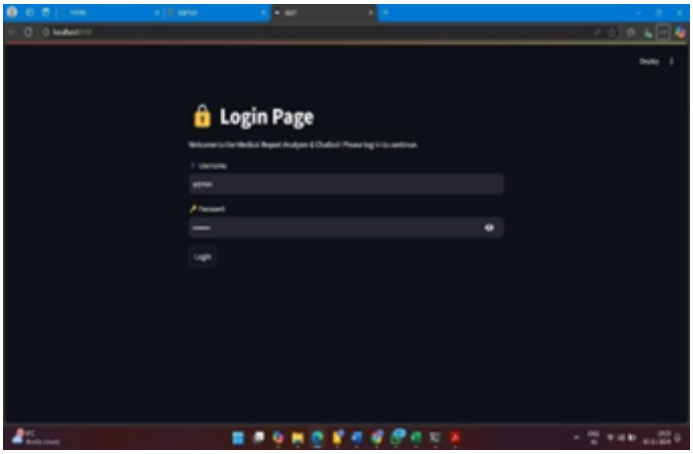

Fig. 3. Login page

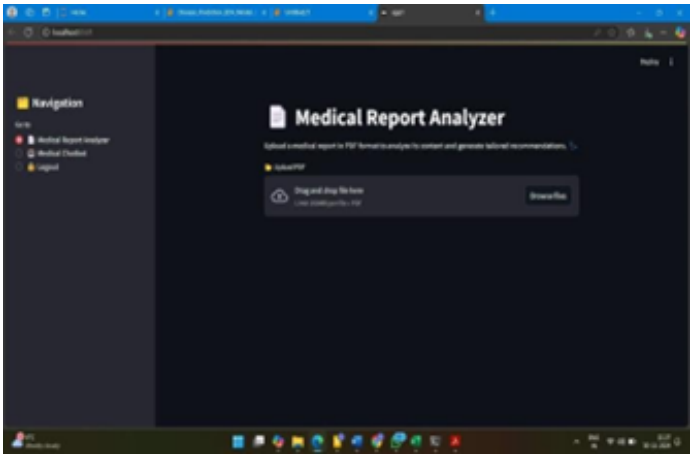

Fig. 4. Navigation Bar

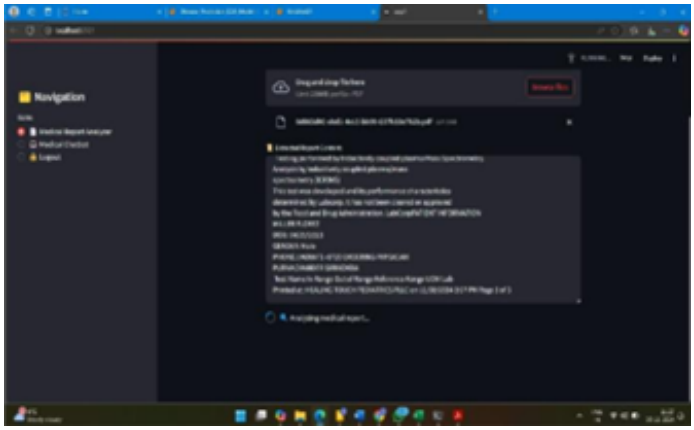

Fig. 5. Uploading the medical report

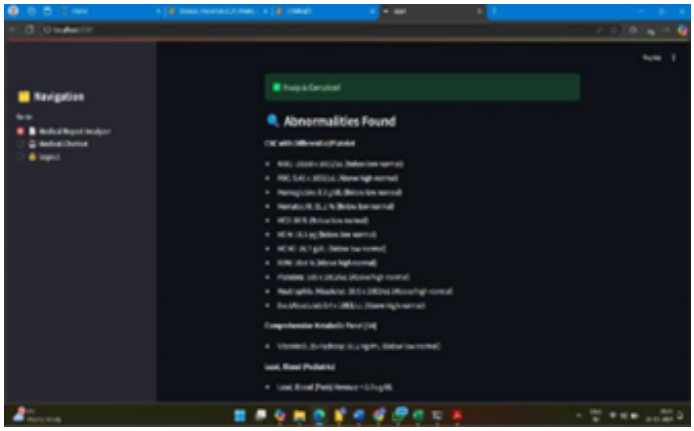

Fig. 6. Output showing abnormalities

In order to perform these functions effectively, the design is complemented by the technique known as the prompt engineering, which involves the generation of specific prompts for the Analysis of the data by the AI. These prompts define the actions of the model that is to define if there are abnormalities in the given data, suggest medications for such, or recommend changes in the patient's lifestyle. Other elements used in the design are text extraction, usage of a chatbot interface, and consideration of failure scenarios including a missing API key or an incorrect login attempt. Such planning makes the app easy to use, fast in response and provides accurate data to fit the user's purposes.

VI. Conclusion

This research work proves how AI can augment the processes of diagnosing diseases during the analysis of the medical reports and images. The system is designed to pull apart and categorise text and medical scans data that assist doctors in identifying early signs of major, such as heart diseases or diabetes. It then gives the patient individual recommendations including drugs to take or changes in one's lifestyle due to individual health information hence more effective in patients treatment.

The increased variety of data such as text and images that the system can facilitate puts it in a vantage position as a tool in today's health sector. The more it collects data the more it learning and the insights it gives are more accurate over time. Apart from the last, this way doctors can reach better decisions quickly, and, therefore, patients get optimal and quicker individualized care which ultimately yields improved patient health and increased availability of quality care.

Streamlined Medical Report Analysis where he application allows users to upload medical reports in PDF format, extracts text, and provides tailored analysis with insights on abnormalities, conditions, and recommendations.Custom Recommendations Using Generative AI, the app suggests medications, supplements, and activities based on report analysis, providing detailed explanations for each recommendation.Interactive Chatbot Functionality Users can interact with a chatbot to discuss their reports or health-related concerns, with responses generated from analyzed data or general medical knowledge. Secure and Accessible Platform where app includes a login system for secure access, with intuitive navigation for users to explore analysis, chatbot, and logout options effectively.

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